

EK SATHI

AI-Powered Student Companion for Learning, Projects, Career & Opportunities

1. Project Introduction

Ek Sathi ek cloud-based AI-powered student companion platform hai, jiska main objective students ke learning, project development, career planning, opportunities aur technical knowledge ko ek single intelligent platform par provide karna hai.

Aaj students ko different requirements ke liye multiple platforms use karne padte hain—AI assistants, GitHub, internship websites, hackathon platforms, resume tools, technology news websites aur productivity applications.

Ek Sathi in activities ko ek unified AI-powered ecosystem mein combine karega.

Platform ka central component ek **AI Chat Assistant** hoga, jiske saath student normal conversation ke through apne questions, projects, studies, career aur planning related tasks handle kar sakega.

Main chat ke alawa platform par **5 major feature cards** honge, jinke through specialized functionality available hogi.

2. Project Objective

Ek Sathi ka primary objective ek aisa AI assistant develop karna hai jo:

- Students ko personalized assistance provide kare.
 - Student ke goals aur preferences ko remember kare.
 - Projects aur hackathons ke development mein help kare.
 - Public GitHub repositories aur websites ko study material mein convert kare.
 - Resume aur career planning mein assistance de.
 - Relevant internships aur hackathons discover kare.
 - Technology-related latest information provide kare.
 - Student ke learning aur scheduling tasks ko chat ke through manage kare.
 - Multiple cloud services aur AI APIs ko integrate karke real-world cloud application demonstrate kare.
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3. Application Structure

Ek Sathi ko broadly do parts mein divide kiya jayega:

A. Main AI Chat Interface

Ye application ka primary interface hoga.

B. Five Feature Cards

Specialized functionalities ke liye:

1. **Project & Hackathon Lab**
2. **GitHub & Website Study**
3. **Career & Resume**
4. **Opportunity Finder**
5. **Tech News**
6. **My Memory**

Note: Technically yahan **6 feature cards** hain. Main chat interface inke alawa central interface hoga.

4. Main AI Chat Interface

Main chat Ek Sathi ka central intelligence interface hoga.

Student is interface ke through directly AI ke saath interact karega.

Chat system ko general-purpose aur context-aware banaya jayega.

Main Chat Capabilities

- General student assistance
- Technical questions
- Programming assistance
- Concept explanation
- Study assistance
- Project discussion
- Career discussion
- Hackathon discussion
- Task planning
- Study planning
- Scheduling

- Personalized recommendations
- Memory-based conversations
- News discussion

Study planner aur scheduling ko separate feature card nahi banaya jayega. Ye functionality **main chat ke andar naturally available** hogi.

5. Feature Card 1 — Project & Hackathon Lab

Ye module students ko projects aur hackathons ke complete development lifecycle mein assist karega.

Project Assistance

System help karega:

- Project idea generation
- Idea validation
- Problem statement analysis
- Feature planning
- Technology stack selection
- System architecture
- Database planning
- API planning
- MVP definition
- Development roadmap
- Task breakdown
- Documentation
- Presentation preparation

Hackathon Assistance

System provide karega:

- Hackathon problem statement analysis
- Solution ideation
- Idea comparison
- Technical feasibility analysis
- Technology selection
- MVP planning
- Development strategy
- Team task distribution
- Timeline planning
- Pitch preparation
- Presentation assistance
- Mock judging

- Technical and conceptual questions preparation

Project aur hackathon ko ek hi module mein rakhne ka reason ye hai ki dono ka workflow largely connected hai—hackathon ke andar bhi student ko idea se lekar project development aur final presentation tak assistance chahiye hoti hai.

6. Feature Card 2 — GitHub & Website Study

Ye Ek Sathi ka ek major learning-focused feature hoga.

Is module ke through student kisi **public GitHub repository ya website URL** ko provide kar sakega.

System available information ko analyze karke us resource ko interactive study environment mein convert karega.

6.1 GitHub Repository Study

Public GitHub repository ko analyze karte waqt system available information ko process karega, jaise:

- Repository metadata
- README
- Folder structure
- Source code
- Configuration files
- Dependencies
- Documentation
- Public project information

AI repository ko understand karke student ko explain kar sakega:

- Project ka overall purpose
- Architecture
- Technology stack
- Folder structure
- Important files
- Code flow
- Data flow
- Functions
- Classes
- APIs
- Database integration
- Authentication
- Dependencies
- Deployment configuration

Student repository ke context mein further questions bhi ask kar sakega.

Iska purpose sirf code explain karna nahi, balki **existing software projects ko learning material mein convert karna** hai.

6.2 Website Study / Website Decoder

GitHub ke alawa student kisi public website ka URL bhi provide kar sakega.

System website ko jitna technically accessible aur legally processable information ke basis par decode kar sakega, utna analyze karega.

Possible analysis:

- Website structure
- Technology identification
- Frontend framework
- Backend indicators
- APIs
- Public assets
- Page structure
- Scripts
- Libraries
- Metadata
- Public configuration information
- SEO structure
- Publicly accessible technical information

AI collected information ko organize karke website ka technical overview provide karega.

Important Limitation

System private information, authentication-protected resources, private APIs, credentials, secrets ya restricted content ko access ya extract nahi karega.

Is feature ka purpose **technical learning and analysis** hoga, unauthorized access nahi.

7. Feature Card 3 — Career & Resume

Career & Resume module students ke career development ko manage karega.

Ismein do major components honge:

- Resume Analyzer
 - Career Planner
-

7.1 Resume Analyzer

Student apna resume upload karega.

System resume ko analyze karke provide karega:

- Resume structure analysis
- Skills identification
- Project analysis
- Experience analysis
- Strengths
- Weaknesses
- Missing information
- ATS-related analysis
- Keyword analysis
- Formatting suggestions
- Improvement recommendations

Student kisi particular job role ke according bhi resume ko analyze karwa sakega.

7.2 Career Planner

Student apna career goal define karega.

System student ke:

- Current skills
- Education
- Experience
- Projects
- Interests
- Career goal
- Current knowledge level

ko consider karke personalized career roadmap generate karega.

Roadmap mein include ho sakta hai:

- Required skills

- Learning sequence
- Technologies
- Projects
- Certifications
- Practice requirements
- Internship preparation
- Interview preparation

Career roadmap static nahi hoga. Student ke progress aur changing goals ke according ise update kiya ja sakega.

8. Feature Card 4 — Opportunity Finder

Opportunity Finder students ke liye relevant opportunities discover aur organize karega.

System multiple sources se opportunities collect karne ke liye APIs, feeds aur permitted/public data sources ka use karega.

Opportunity Types

- Internships
 - Hackathons
 - Scholarships
 - Competitions
 - Conferences
 - Workshops
 - Open-source programs
 - Student programs
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Opportunity Processing

Collected opportunities ko system:

1. Clean karega
2. Duplicate remove karega
3. Categorize karega
4. Eligibility information identify karega
5. Required skills identify karega
6. Deadline extract karega
7. Relevance determine karega

User profile ke basis par AI relevant opportunities ko prioritize karega.

Opportunity Information

- Opportunity name
- Organization
- Type
- Description
- Eligibility
- Required skills
- Deadline
- Location
- Remote/On-site
- Application information
- Source link

Iska objective student ko sirf opportunities ki list dena nahi, balki **relevant opportunities identify karna** hai.

9. Feature Card 5 — Tech News

Tech News module students ko important technology updates ke saath connected rakhega.

System multiple permitted sources aur APIs se technology-related information collect karega.

Major Categories

- Artificial Intelligence
 - Machine Learning
 - Cloud Computing
 - Cybersecurity
 - Programming
 - Web Development
 - DevOps
 - Open Source
 - Databases
 - Developer Tools
 - Startups
 - Emerging Technologies
-

News Processing Pipeline

Raw information directly user ko show nahi ki jayegi.

Pipeline:

```
graph TD; Sources --> DataCollection[Data Collection]; DataCollection --> ContentCleaning[Content Cleaning]; ContentCleaning --> DuplicateDetection[Duplicate Detection]; DuplicateDetection --> Categorization; Categorization --> RelevanceAnalysis[Relevance Analysis]; RelevanceAnalysis --> AISummarization[AI Summarization]; AISummarization --> Database; Database --> PersonalizedNewsFeed[Personalized News Feed];
```

AI important information identify karke news ko concise aur understandable format mein present karega.

Interactive News Discussion

News module ka important part interactive discussion hoga.

Student kisi particular news topic ko select karke uske context mein AI conversation start kar sakega.

AI article/topic ke available context ko use karke:

- Explanation
- Technical breakdown
- Impact analysis
- Related technologies
- Learning direction

provide kar sakega.

Isse Tech News ek simple news feed nahi, balki **technology learning platform** ka part ban jayega.

10. Feature Card 6 — My Memory

My Memory Ek Sathi ki personalization layer hogi.

AI student ki important aur relevant information ko long-term context ke liye store kar sakega.

Possible Memory Categories

- Career goals
- Current projects
- Technical skills
- Learning interests
- Preferred technologies
- Learning preferences
- Communication preferences
- Long-term goals
- Important plans

AI ko user ki complete conversation history har request mein send nahi karni hogi.

Instead, relevant memories retrieve karke AI context mein provide ki jayengi.

Memory Management

Student ko complete control diya jayega:

- View memories
- Add memories
- Edit memories
- Delete memories
- Manage personalization

AI ko har conversation permanently remember nahi karni chahiye.

Sirf useful and meaningful information ko long-term memory mein store kiya jayega.

11. Personalized AI Behaviour

Ek Sathi ka objective user ki talking style aur preferences ke according gradually personalize hona hai.

Iske liye unrestricted model training ki jagah **structured user preferences + memory + conversation context** ka use kiya jayega.

System user ke preferred:

- Language
- Response style
- Explanation depth

- Technical level
- Communication style

jaise preferences ko maintain kar sakta hai.

Isse AI future conversations mein more personalized responses provide karega.

12. Study & Scheduling Through Chat

Study scheduling ka separate card nahi hoga.

Student main chat ke through:

- Study plan
- Daily tasks
- Weekly plan
- Revision plan
- Project deadlines
- Hackathon deadlines
- Learning schedule
- Priority management

handle kar sakega.

AI user ke available time, deadlines aur goals ke according plan create kar sakta hai.

Future versions mein external calendar integration bhi add ki ja sakti hai.

13. Technology Stack

Frontend

Next.js

Use:

- Web application
- Chat interface
- Dashboard
- Feature cards
- Routing
- UI components
- Client-side interactions

Frontend Deployment

Vercel

Use:

- Next.js hosting
- Production deployment
- Continuous deployment
- Environment configuration
- Scalable frontend delivery

14. Backend

FastAPI

FastAPI Python-based backend ke liye use kiya jayega.

Backend handle karega:

- AI requests
- Authentication verification
- Database operations
- GitHub analysis
- Website analysis
- Resume processing
- Career analysis
- Opportunity processing
- News processing
- Memory operations
- External API communication
- Application business logic

15. LLM Layer

OpenRouter

OpenRouter ko LLM gateway ke roop mein use kiya jayega.

LLM ka use:

- Main AI conversation
- Project assistance
- Hackathon assistance
- GitHub analysis
- Website analysis
- Resume analysis
- Career planning
- Opportunity relevance analysis
- News classification
- News summarization
- Memory extraction

ke liye hoga.

Task ke according different models use karna possible hoga.

16. Firebase

Firebase ka primary role **authentication and identity management** hoga.

Possible uses:

- User registration
- Login
- Google authentication
- Email authentication
- Authentication state
- User identity management

Firebase ko authentication ke liye primarily use karne se architecture comparatively clean rahega.

17. Supabase

Supabase PostgreSQL-based structured data storage ke liye use kiya ja sakta hai.

Possible tables/entities:

- Users/Profile
- Conversations
- Messages
- Memories

- Projects
- Roadmaps
- Opportunities
- News
- Repositories
- Resume Analysis
- Schedules

Supabase application ke structured data ko manage karega.

18. Cloudinary

Cloudinary uploaded files aur media ke management ke liye use kiya jayega.

Possible uses:

- Resume files
- Profile images
- Project media
- Other supported documents

Sensitive files ke liye secure access and authorization implement ki jayegi.

19. Memory / Fact Storage API

Ek dedicated memory/fact-storage service ya API ko long-term user facts ke liye integrate kiya jayega.

Iska purpose:

- Important user facts store karna
- Relevant facts retrieve karna
- User personalization maintain karna
- AI context ko optimize karna

Memory service ko main application database se logically separate rakha ja sakta hai agar architecture ke liye beneficial ho.

20. GitHub Integration

GitHub integration ka use:

- Public repository information
- Repository metadata
- README
- File structure
- Public source code
- Repository analysis

ke liye kiya jayega.

Large repositories ke case mein complete repository ko ek hi LLM request mein bhejne ke bajay relevant files/context ko process karna hoga.

21. Website Analysis Engine

Website Study ke liye backend ek analysis pipeline maintain karega.

Basic architecture:

```
Website URL
  ↓
URL Validation
  ↓
Public Content Collection
  ↓
HTML / Metadata Processing
  ↓
Asset & Script Analysis
  ↓
Technology Detection
  ↓
Content Structuring
  ↓
LLM Analysis
  ↓
Interactive Study Context
```

System sirf publicly accessible aur permitted information ko process karega.

22. News Collection System

Tech News ke liye automated collection system create kiya jayega.

Ismein:

- Scheduled data collection
- Source management
- Content extraction
- Cleaning
- Deduplication
- Categorization
- AI summarization
- Relevance scoring
- Database storage

include honge.

Background jobs ke through news pipeline periodically run ho sakti hai.

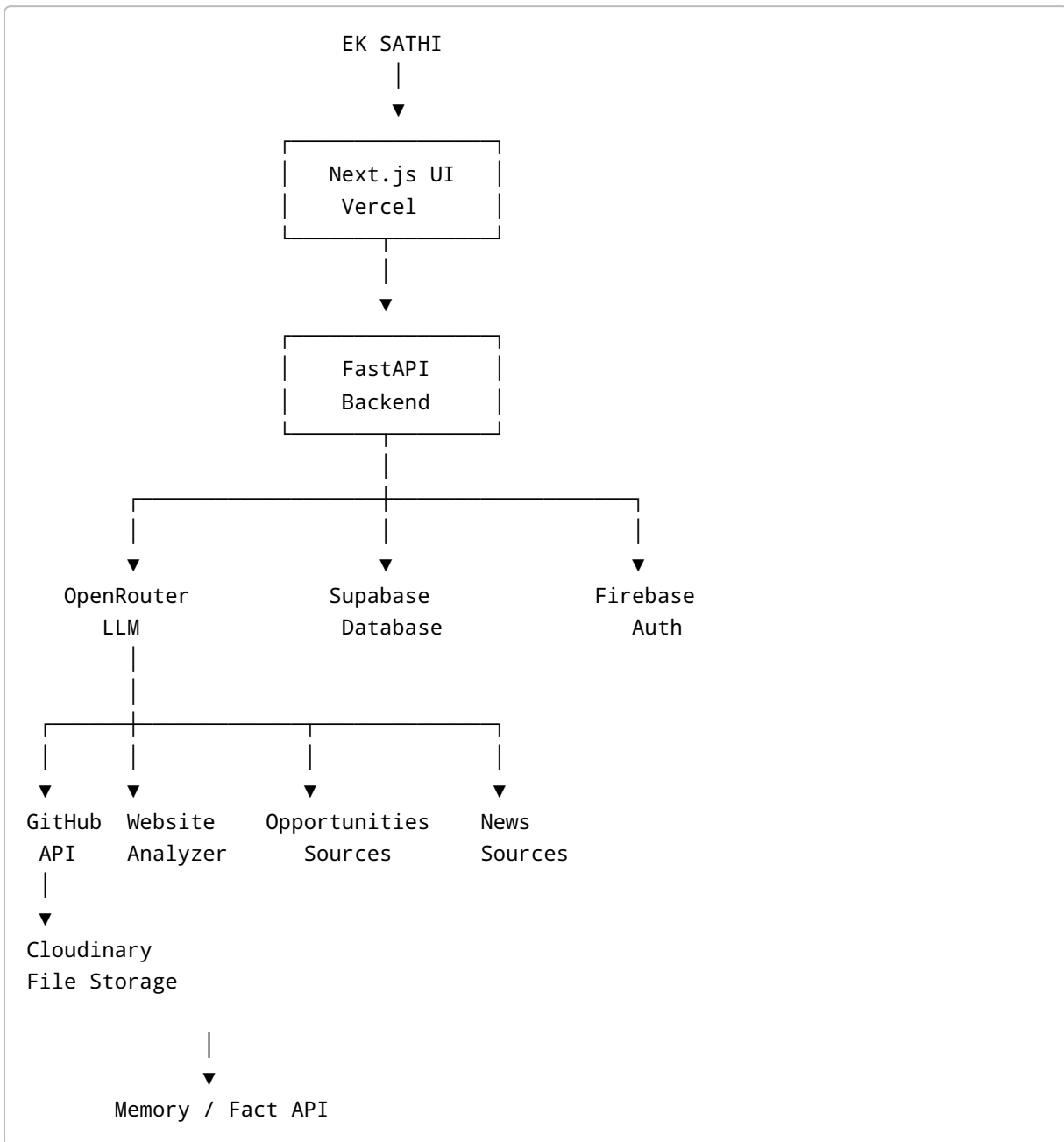
23. Opportunity Collection System

Opportunity Finder bhi automated data pipeline use karega.

```
graph TD; A[Opportunity Sources] --> B[Collection]; B --> C[Cleaning]; C --> D[Duplicate Removal]; D --> E[Classification]; E --> F[Eligibility Extraction]; F --> G[Deadline Extraction]; G --> H[Relevance Scoring]; H --> I[Database];
```

Isse student ko manually multiple websites search karne ki requirement kam hogi.

24. Overall System Architecture



25. AI Chat Context Architecture

AI response generate karte waqt system sirf current message ko LLM ke paas nahi bhejega.

Context pipeline:

```
User Message
  ↓
Current Conversation
  ↓
Relevant User Profile
  ↓
Relevant Memories
  ↓
Feature-Specific Context
  ↓
System Instructions
  ↓
OpenRouter
  ↓
AI Response
```

Feature-specific context ke examples:

- GitHub Study → Repository context
- Website Study → Website analysis context
- Resume → Resume content
- Career → Career profile
- News → Selected news context
- Project → Project information

Is architecture se AI ko har request par unnecessary information nahi bhejni padegi.

26. Data Storage Structure

Major entities:

```
Users
Profiles
Conversations
Messages
Memories
Projects
Hackathons
Repositories
Website Analyses
Resumes
```

Career Roadmaps
Opportunities
News
Schedules

Har user ka data logically isolated hoga.

27. Security & Privacy

Project mein security beginning se architecture ka part honi chahiye.

Important measures:

- Authentication
- Authorization
- User-level data isolation
- Secure API keys
- Environment variables
- HTTPS
- Input validation
- Rate limiting
- Secure file access
- Database security rules
- Memory management
- Delete-data functionality

API keys ko frontend code mein expose nahi kiya jayega.

28. Cloud Computing Concepts Demonstrated

Ek Sathi Cloud Computing project ke perspective se multiple concepts demonstrate karega:

Cloud Deployment

Application cloud infrastructure par deployed hogi.

Cloud Authentication

Firebase Authentication.

Cloud Database

Supabase/PostgreSQL.

Cloud Storage

Cloudinary.

AI as a Service

OpenRouter LLM APIs.

API-Based Architecture

Multiple external services communicate karenge.

Distributed Services

Authentication, database, storage, AI aur data collection different services ke through handle honge.

Background Processing

News aur opportunity collection automated jobs ke through run ki ja sakti hai.

Scalability

Application ke different components ko independently scale kiya ja sakta hai.

29. Development Phases

Phase 1 — Core Platform

- Next.js frontend
 - FastAPI backend
 - Firebase authentication
 - Supabase database
 - OpenRouter integration
 - Main AI chat
 - Basic conversation history
 - Basic memory
 - Vercel deployment
-

Phase 2 — Core Student Features

- Project & Hackathon Lab
- GitHub Study
- Website Study

- Resume Analyzer
 - Career Planner
 - Career Roadmap
-

Phase 3 — External Intelligence

- Opportunity Finder
 - Internship collection
 - Hackathon collection
 - Tech News
 - News processing pipeline
 - Personalized recommendations
-

Phase 4 — Advanced Personalization

- Advanced memory
 - User preference learning
 - Chat-based scheduling
 - Study planning
 - Notifications
 - Calendar integration
 - Better personalization
-

30. Final Feature Structure

Main Interface

AI Chat

General-purpose personalized AI assistant.

Chat ke through:

- Study
 - Coding
 - Planning
 - Scheduling
 - Project discussion
 - Career discussion
 - General assistance
-

Six Feature Cards

1. Project & Hackathon Lab

Project creation, development aur hackathon preparation.

2. GitHub & Website Study

Public GitHub repositories aur websites ko technically analyze karke interactive study environment.

3. Career & Resume

Resume analysis, career planning aur personalized roadmap.

4. Opportunity Finder

Internships, hackathons aur other student opportunities.

5. Tech News

Technology news collection, filtering, summarization aur interactive discussion.

6. My Memory

AI ki long-term user memory aur personalization management.

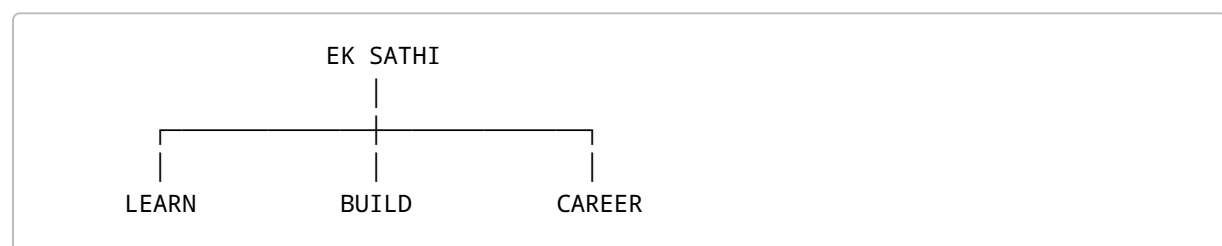
31. Final Vision

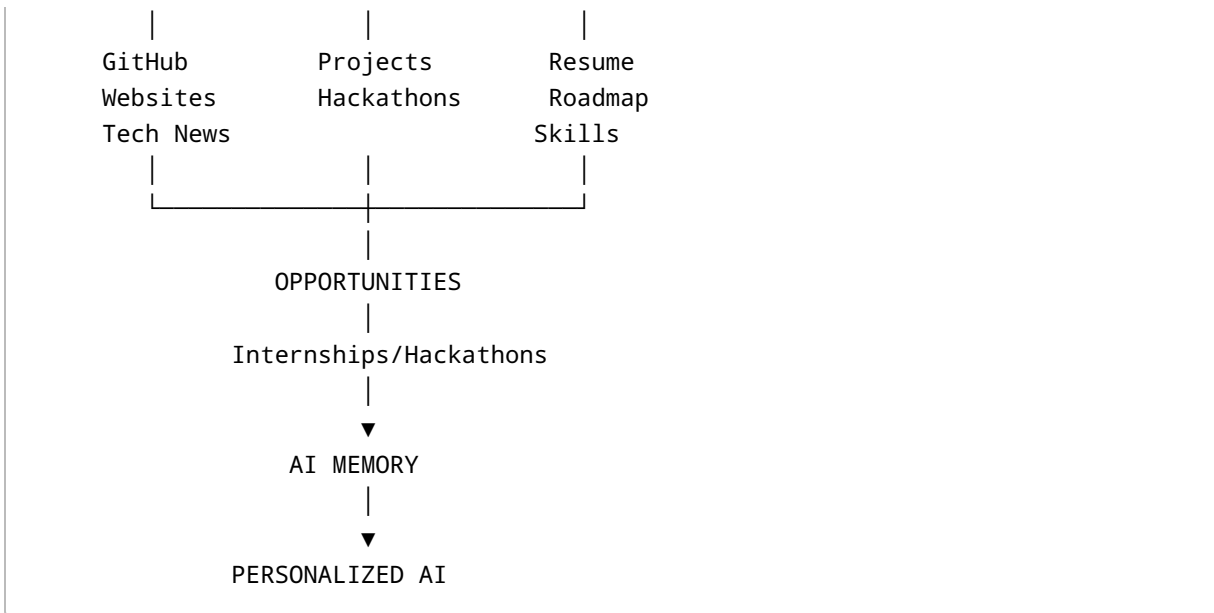
Ek Sathi ka vision ek normal AI chatbot se kaafi bada hai.

Is project ka objective ek **personal AI student companion** develop karna hai jo student ke learning aur career journey ke different stages mein continuously assist kare.

Platform ka central AI student ke context ko understand karega, important information remember karega aur specialized modules ke through learning, projects, career aur opportunities ko support karega.

The complete ecosystem can be represented as:





32. Conclusion

Ek Sathi ek cloud-native AI-powered student companion platform hoga jo conversational AI, personalized memory, project assistance, hackathon preparation, GitHub and website study, resume analysis, career planning, opportunity discovery aur technology intelligence ko ek single ecosystem mein combine karega.

Project mein **Next.js, FastAPI, Firebase, Supabase, Cloudinary, Vercel, OpenRouter aur external APIs/ data sources** ko integrate karke ek practical cloud architecture develop ki jayegi.

Is project ka sabse important aspect ye hoga ki Ek Sathi sirf user ke questions ka answer nahi karega. Ye user ke **goals, skills, projects, interests aur preferences** ko context mein lekar progressively more personalized assistance provide karega.

Long-term vision ke according, Ek Sathi ko ek **AI-powered Student Operating System** ke form mein develop kiya ja sakta hai—jahan student apni learning, projects, career planning, opportunities aur technical knowledge ko ek intelligent platform se manage kar sake.

EK SATHI

Learn. Build. Prepare. Grow.

Your AI Companion for the Student Journey.